

Avneesh Kumar

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LinkedIn — LeetCode — GFG

PROFILE

Aspiring AI Engineer with strong foundation in Python, DSA, and backend systems. Experienced in building real-time AI pipelines, computer vision applications, and scalable APIs. Interested in ML/DL, cloud deployment, and production-level AI systems.

EDUCATION

B.Tech – Computer Science & Engineering, University of Lucknow	2022–2026 — CGPA: 7.50
XII – Army Public School	2022 — 93%
X – Army Public School	2020 — 85%

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, C, C++
Core CS:	DSA, OOP, DBMS, Operating Systems
AI/ML:	Computer Vision, Video Processing, Model Inference, NLP Basics
Frameworks:	FastAPI, Node.js, Express.js, React.js
Cloud/DevOps:	Basic AWS, Docker (familiar), REST APIs, CI/CD concepts
Databases:	MongoDB, MySQL
Tools:	Git, GitHub, Postman, VS Code

KEY PROJECTS

Secure-Vision: Real-Time Violence Detection System GitHub

Python, FastAPI, WebSockets, Computer Vision, AI Pipelines

Developed a real-time AI-powered surveillance system using VideoMAE models for violence detection and crime classification.

Designed and implemented an end-to-end AI inference pipeline including frame extraction, preprocessing, batching, and model prediction.

Built scalable backend using FastAPI with WebSockets for live streaming and SSE for asynchronous video processing.

Optimized inference using frame skipping, buffering (16-frame window), and threshold-based alert triggering.

Integrated automated report generation and alert system with cooldown logic and detection history tracking.

AI Code Review Platform Live

React.js, Node.js, Gemini API

Developed AI-driven code analysis tool providing real-time feedback on syntax, structure, and performance.

Integrated LLM APIs and built modular backend services for scalable request handling.

Designed responsive frontend for seamless user interaction and efficient code evaluation workflows.

Hostel Management System (HMS) Live

React.js, Node.js, MongoDB, JWT

Built full-stack hostel management system with role-based authentication and secure access control.

Implemented OTP verification, automated eligibility checks, and optimized API endpoints.

Improved backend performance and ensured data consistency across modules.

EXPERIENCE

Web Development Intern – VaultofCodes.in June–July 2025

Developed full-stack features using React.js and Node.js, improving application functionality and performance.

Built REST APIs and optimized backend logic for better response time and scalability.

Collaborated with team members on UI improvements, debugging, and deployment workflows.

AICTE Virtual Internship – Front End Web Development 2025

Built responsive web applications using React.js and Tailwind CSS.

Strengthened fundamentals of UI development, API integration, and version control.

Student Coordinator – Incubation Cell 2023–2024

Organized technical events and collaborated with teams to manage operations and outreach.

Mentored students and contributed to community-driven tech initiatives.

PROBLEM SOLVING

- Solved 300+ DSA problems across LeetCode and GeeksforGeeks.
- Strong in recursion, dynamic programming, trees, and graphs.
- Regular coding contest participant.

CERTIFICATIONS

- Deloitte Cybersecurity & Data Analytics Simulation – Forage
- Google Cloud ML Fundamentals (Basics)